

LOURDES MATHA
COLLEGE OF SCIENCE & TECHNOLOGY
Bureau Veritas Certified ISO 9001-2015 Institution
Managed by Archdiocese of Changanacherry

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Workbook

20MCA283-Deep Learning

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DEEP LEARNING (20MCA283)

MODULE-I

PART –A (3 marks each)

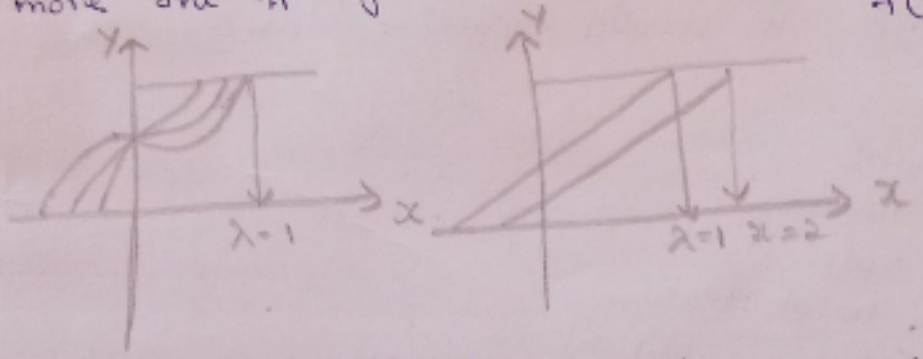
1. Describe sigmoid activation functions.

Sigmoid fn :- widely used in back propation network because of the relationship between the value of the fn at a point and value of derivative at that point which reduces the computational burden during training.

Two Type : (i) Binary Sigmoid function
(ii) Bipolar Sigmoid function

(i) Binary Sigmoid fn, activation fn performs input editing btw 0 & 1. It is positive in nature. It is always bounded which means its o/p cannot be less than 0 and more than 1. It is also strictly increasing in nature which means more the i/p higher would be the output

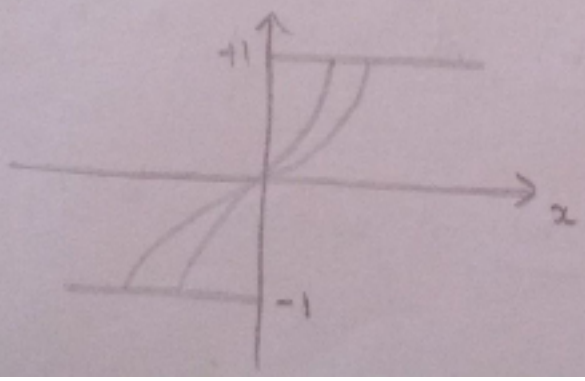
$$f(x) = sigm(x) = \frac{1}{1+\exp(-x)}$$



(ii) Bipolar sigmoidal fn, activation fn performs input editting btw -1 and 1. It can be positive or negative in nature. It is always bounded which means its o/p cannot be less than -1 and more than 1. It is also strictly increasing in nature like sigmoid function.

$$f(x) = Sigm(x) = \frac{2}{1+\exp(-x)} - 1$$

$$= \frac{1-\exp(x)}{1+\exp(x)}$$



2 Write the gradient descent algorithm.

Gradient descent algorithm is iteratively calculate the next point using gradient of the current position. Then scalar it by a learning rate and subtract the obtained values from the current position. 'η'(eta) learning rate it scales the gradient and thus control the step size when eta became smaller than longer is gradient descent converger.

Step 1: Choose a stating point (initialization).
Step 2: Calculate gradient at this point
Step 3: Make a scaled step in the opposite direction to the gradient (minimize)
Step 4: Repeat step 2 and step 3 until one of the criteria.
Step 5: Maximum no:of iteration is reached.
Step 6: Step is smaller than the tolerance.

Parameter

* Starting point
* Gradient function
* Learning rate
* Maximum no:of itration
* Tolerence (stoping condition)

Algorithm

initialize w, b;
iterate over data;
compute y'
compute L(w, b)

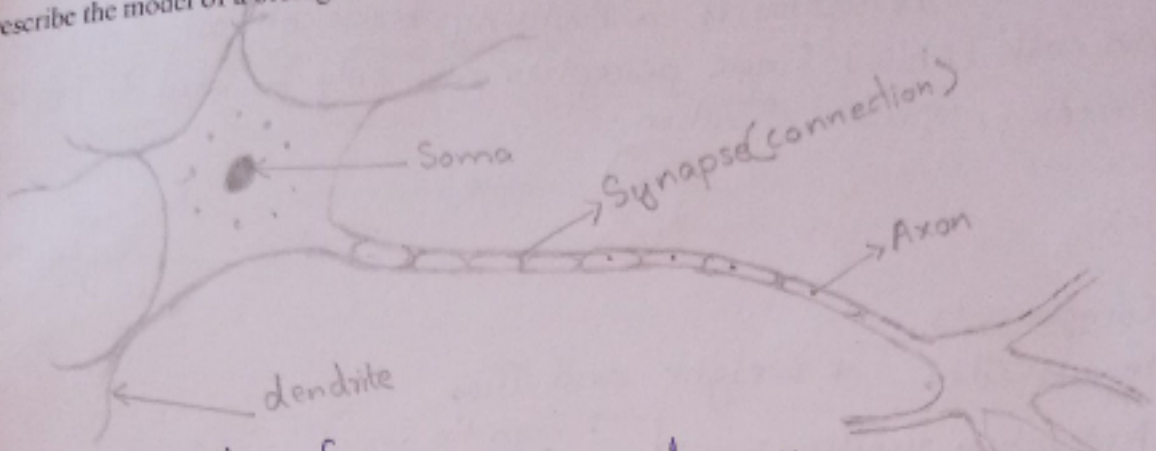
$w_{t+1} = w_t - \eta \Delta w_t$

$b_{t+1} = b_t - \eta \Delta b_t$

where, $\Delta w_t = \frac{\partial L(w,b)}{\partial w}$, at $w = w_t$, $b = b_t$

$\Delta b_t = \frac{\partial L(w,b)}{\partial b}$, at $w = w_t$, $b = b_t$

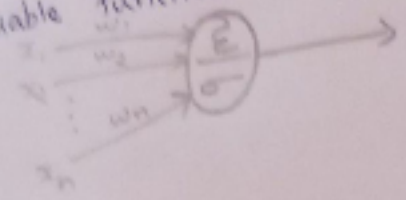
3. Describe the model of a biological neuron.



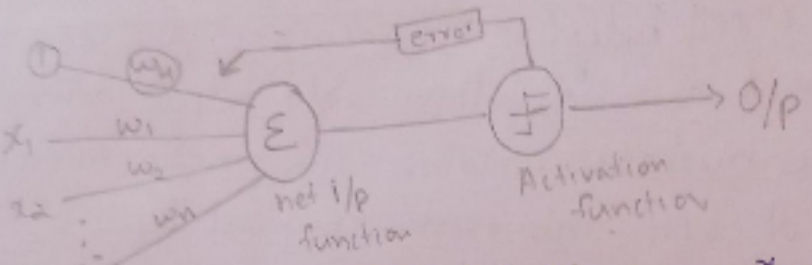
→ The majority of neurons encode their activation or output as a series of brief electrical pulses.
→ The neuron's cell body (soma) processes the incoming activation and converts them into output activation.
→ The neuron's nucleus contains the genetic material in the form of DNA. This exists in most types of cells not just neurons.
→ Dendrites are fibres which emanate from the cell body and provide the receptive zones the receive activation from other neurons.
→ Axons are fibres acting as transmission ling that send activation to other neurons.
→ Junctions that allow signal transmission between the axon and dentrites called synapse.
→ The process of transmission is by diffusion of chemicals called neuro transmitters across the synaptic cleft.
→ Center part is called nucleus
→ It is basic model of biological neuron.

4. Explain Perceptron learning algorithm.

Perceptron is a building block of an Artifical neural network (ANN). Single perceptron can only be used to implement linearly seperable function.



Components
* Inputnodes
* weight and Bias
* Activation functions → sign fn, stip fn, sigmoid



step1 : $\sum w_i * x_i = x_1 w_1 + x_2 w_2 \cdots x_n w_n$

$\sum w_i x_i + b$

step 2 : $Y = f(\sum w_i * x_i + b)$.

It is 2 steps of perceptron.

Perceptron is a machine learning algorithm for supervised learning of binary classifiers. In perception the weight coefficient is automatically learned. The activation function applies a step rule to check whether the weight function greater than zero.

Algorithm

P ← i/ps with label 1 ;
N ← i/ps with label 0 ;
initialize w randomly ;
while !convergence do
pick random x ∈ P U N ;
if x ∈ P and w . x < 0 then
w = w + x ;
end
if x ∈ N and w . x ≥ 0 then
w = w - x ;
end
end